

Stakeholder Engagement Plan: Optimizing RAG Document Retrieval for Agronomic Advice -TEAM BOSUMTWI

Stakeholder Group 1: Agricultural Extension Workers

1. Identification

Who they are: These are the boots on the ground local and state government officials, NGO, or community professionals who travel directly to rural farms to diagnose issues and advise farmers.

Why they are important: They are the actual users of our retrieval system. If they don't trust the documents the system pulls up, the tool will sit unused, and the project fails.

What matters most to them: Getting accurate, fast, and scientifically backed answers. They need to protect their professional credibility and avoid giving advice that could harm the communities they serve.

2. Potential Benefits and Risks

Benefits: Instead of hauling around heavy physical crop manuals, they get a rapid decision-support tool that pulls the exact evidence they need while standing right in the field.

Harms or Concerns: They rightly worry the system might retrieve irrelevant info or hallucinate, causing them to give bad advice. There is also an underlying anxiety that AI might eventually be used to replace their jobs.

Considerations: Their social capital and reputation in the community are everything. Ethically, they are the ones taking responsibility for the advice given to the farmer.

3. Engagement Method

Method: Participatory Design (from the /Catalogue)

Why it's appropriate & suitable: Extension workers know the chaos of their daily jobs better than we do. Co-designing the interface ensures it actually fits their field workflow rather than a sterile lab environment.

How it would be conducted: We will run collaborative workshops where they map out a typical day using design workbooks. We'll hand them prototype screens so they can tell us exactly where they need source citations, warnings, and relevance scores placed to feel confident using the tool.

Stakeholder Group 2: Smallholder Farmers in West Africa

1. Identification

Who they are: The rural farmers actually working the land. They often have limited digital literacy and describe problems based on what they see (e.g., "yellow spots on leaves") rather than using scientific terms.

Why they are important: They are the ultimate beneficiaries. Their natural, conversational way of describing crop issues is exactly what our retrieval system has to decode and bridge.

What matters most to them: Saving their harvest. They need actionable, affordable, and locally available solutions to protect their livelihoods.

2. Potential Benefits and Risks

Benefits: Faster, more accurate diagnoses mean less crop loss, better yields, and more stable income for their families.

Harms or Concerns: If the system misunderstands their local phrasing, it might recommend the wrong pesticide dosage, destroying their crop and poisoning their soil.

Considerations: We must respect their traditional knowledge and ensure that retrieved interventions do not rely on expensive, inaccessible commercial farming tools.

3. Engagement Method

Method: Deliberative Workshops / Mini-publics (from the Action Catalogue)

Why it's appropriate & suitable: Digital surveys won't work here. A facilitated, spoken workshop in their local language respects their oral traditions and creates a comfortable space to share how they actually talk about crop diseases.

How it would be conducted: We will hold in-person community gatherings in rural hubs. Facilitators will present mock crop failure scenarios, and farmers will discuss how they'd describe the symptoms. We will then test if the solutions our system pulls up actually make sense and feel actionable to them.

4. Influence on Project Design

Dataset: The local phrases and symptom descriptions we collect from the farmers' workshops will go straight into our training queries to help fix the vocabulary mismatch problem.

Model Design: The extension workers' feedback will dictate our metadata rules—the system must return verifiable source links and publisher names alongside every text chunk, or it won't be trusted.

Outputs: Based on what the extension workers flag as most dangerous, we'll tweak the UI to boldly highlight critical constraints (e.g., "Toxic to local water sources").

Evaluation: We will redefine our relevance labels. A "Highly Relevant" (Score 3) document won't just mean it is scientifically accurate; it must mean it is practically actionable for a smallholder.

Deployment: If we find out during the workshops that internet access in the field is too spotty, we will pivot to deploying this as a low-bandwidth or offline-first application.